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RESEARCH ARTICLE

Value of circulating tumor cell assisting low-dose computed tomography in screening pulmonary nodules based on existing liquid biopsy techniques: a systematic review with meta-analysis and trial sequential analysis

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Abstract

Objective This study aims to assess the diagnostic utility of circulating tumor cells (CTCs) in conjunction with low-dose computed tomography (LDCT) for differentiating between benign and malignant pulmonary nodules and to substantiate the foundation for their integration into clinical practice.

Methods A systematic literature review was performed independently by two researchers utilizing databases including PubMed, Web of Science, The Cochrane Library, Embase, and Medline, to collate studies up to September 15, 2023, that investigated the application of CTCs in diagnosing pulmonary nodules. A meta-analysis was executed employing Stata 15.0 and Revman 5.4 to calculate the pooled sensitivity, specificity, positive and negative likelihood ratios (PLR and NLR), diagnostic odds ratio (DOR), and the area under the receiver operating characteristic curve (AUC). Additionally, trial sequential analysis was conducted using dedicated TSA software.

Results The selection criteria identified 16 studies, encompassing a total of 3409 patients. The meta-analysis revealed that CTCs achieved a pooled sensitivity of 0.84 (95% CI 0.80 to 0.87), specificity of 0.80 (95% CI 0.73 to 0.86), PLR of 4.23 (95% CI 3.12 to 5.72), NLR of 0.20 (95% CI 0.16 to 0.25), DOR of 20.92 (95% CI 13.52 to 32.36), and AUC of 0.89 (95% CI 0.86 to 0.93).

Conclusions Circulating tumor cells demonstrate substantial diagnostic accuracy in distinguishing benign from malignant pulmonary nodules. The incorporation of CTCs into the diagnostic protocol can significantly augment the diagnostic efficacy of LDCT in screening for malignant lung diseases.

Keywords CTC · Pulmonary nodules · Benign and malignant · Meta-analysis · Trial sequential analysis

Introduction

Cancer stands as the principal cause of mortality worldwide, significantly impeding efforts to enhance global life expectancy. Among cancers, lung cancer remains a leading cause of cancer-related mortality globally, with more than 2.2 million new cases and 1.7 million deaths annually [1]. Thus, the identification and the prevention of lung cancer are crucial for its effective management.

Screening and early diagnosis are established as efficacious strategies for decreasing mortality rates from lung cancer across diverse demographic groups [2]. The value of low-dose computed tomography (LDCT) in reducing mortality associated with lung cancer is substantiated by major trials such as the National Lung Screening Trial (NLST) in the United States and the Nederlands-Leuvens Longkanker Screenings Onderzoek (NLSO) [3, 4]. Nonetheless, subsequent studies indicate a substantial escalation in benign findings post-LDCT screening, with a positive result, initially observed in 24% of participants and later follow-ups revealing benign findings in 96.4% of cases [3]. Particularly, in non-smoking Asian women, there is a high likelihood of overdiagnosis, where the false-positive rate exceeds

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81%, leading to unnecessary imaging and invasive tests for benign conditions [5]. These interventions can cause undue radiation exposure and physical harm, while non-specific pulmonary nodules identified during screenings may also inflict significant physiological and psychological distress on patients [6]. The considerable rate of overdiagnosis underscores the urgent need for more precise diagnostic techniques to accurately determine the malignancy of pulmonary nodules. Current diagnostic modalities, including imaging studies, tissue biopsies, and serum biomarkers, are utilized to differentiate benign from malignant nodules [7, 8]. However, evidence suggests that FDG PET imaging may fail to reliably identify smaller solid malignant nodules. There is a pressing requirement for the development and integration of more accurate diagnostic strategies to reduce the incidence of unnecessary medical interventions and to improve overall patient outcomes in lung cancer management [9].

The accurate evaluation of pulmonary nodules necessitates the development of minimally invasive yet reliable methods. Among potential technologies, serum biomarkers offer cost-effectiveness, efficiency, and rapidity. Circulating tumor cells (CTCs), which are released into the bloodstream from primary tumors or metastatic sites, are under consideration for their diagnostic potential. Clinical and experimental evidence suggests that CTCs may be released during the early stages of tumor progression, potentially remaining dormant [10]. Consequently, the detection of CTCs in serum holds promise for facilitating early diagnosis and initiating timely treatment interventions [11]. Compared to other biomarkers, CTC detection demonstrates superior accuracy in the assessment of lung nodules, making it a focus of recent diagnostic research [12, 13]. Historically, the clinical adoption of CTC detection was limited by low detection rates [14]. However, recent technological advancements have significantly enhanced the sensitivity and specificity of CTC detection methods. These improvements have augmented the diagnostic value of serum CTC assays in determining the malignancy of pulmonary nodules [15]. Despite these advancements, current studies face limitations due to variability in study designs, detection methods, and types of nodules assessed, which restricts broader clinical application.

This study aims to systematically evaluate the clinical utility of CTCs in the diagnosis of nodule malignancy and provide comprehensive references for their potential integration into clinical practice.

Materials and methods

Background of the readers

This manuscript is intended for physicians involved in the management of malignant pulmonary nodules and researchers dedicated to differentiating between benign and malignant pulmonary nodules.

Search strategy and study selection.

A comprehensive literature search was performed by two independent researchers using five databases: PubMed, Web of Science, The Cochrane Library, Embase, and Medline, covering all relevant articles published up to September 15, 2023. The search utilized specific terms including "pulmonary nodule", "lung nodule", "circulating tumor cell", and "CTC". Further, references cited within these articles were scrutinized to identify additional eligible studies. This systematic review and meta-analysis adhered rigorously to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines. The study was registered with PROSPERO (CRD42023465804), ensuring compliance with established research protocols.

Study selection

Inclusion criteria: (1) Studies evaluating the diagnostic utility of CTCs in distinguishing between benign and malignant pulmonary nodules; (2) Research encompassing both types of nodules; (3) Participants free from recent adjuvant therapy or chemotherapy, with no history of other metastatic cancers before CTC detection; (4) Utilization of fresh serum samples.; (5) Reliance on histological examination as the gold standard for diagnosing pulmonary nodules; (6) Availability of sufficient data to calculate true-positive, false-positive, false-negative, and true-negative rates.

Exclusion criteria: (1) Studies not published in English; (2) Non-clinical literature, including basic research, reviews, conference abstracts, experimental records, and letters to editors; (3) Studies providing insufficient data for constructing a 2*2 contingency table.

Data extraction

Data were meticulously extracted by two independent researchers who then cross-verified the results to ensure accuracy. Discrepancies regarding study inclusion were resolved through consultation with a third researcher. The extracted data included the following elements: (1) Basic

characteristics of the studies, such as the first author's name, the publication date, the study design, and the country of origin; (2) Demographic information of the study samples, detailing patient count, mean age and gender distribution; (3) Specifications of the technology used for identifying CTCs, including the thresholds for CTCs cut-off values and the types of applications utilized; (4) Characteristics of the pulmonary nodules investigated, encompassing both size and type; (5) Sensitivity and specificity metrics derived from the studies.

Quality assessment

The quality of the studies included in this review was rigorously evaluated by two independent reviewers using the QUADAS-2 tool, an evidence-based instrument designed to assess the quality across four key domains: patient selection, index test, reference standard, and flow and timing of the study. This tool facilitates a systematic consideration of potential biases and the applicability of the study findings in clinical practice [16].

Statistical analysis

Data analysis was performed using Stata 15.0, while quality assessment was conducted with Revman 5.4. The initial phase involved assessing the presence of a threshold effect and heterogeneity via the Spearman correlation coefficient. A *P*-value greater than 0.05 in the Spearman correlation coefficient indicated the absence of a threshold effect. Subsequent analyses were conducted to explore non-threshold effect-related heterogeneity.

Heterogeneity across studies was evaluated using the Cochran-Q test, considering both the *P*-value and the I^2 statistic. The absence of significant heterogeneity was concluded if $P > 0.05$ and $I^2 \leq 50\%$. Following this, pooled estimates of sensitivity, specificity, positive likelihood ratio (PLR), negative likelihood ratio (NLR), and diagnostic odds ratio (DOR) were calculated. Conversely, if $P \leq 0.05$ or $I^2 > 50\%$, subgroup analyses were undertaken to identify potential sources of heterogeneity. Covariates included in these analyses were: (i) size of lung nodules, (ii) type of lung nodules, (iii) techniques for CTC detection, and (iv) applications of CTCs. A covariate was considered a source of heterogeneity if the subgroup analysis resulted in $P > 0.05$ and $I^2 \leq 50\%$.

The pooled estimates of sensitivity, specificity, PLR, NLR, and DOR, along with their respective 95% confidence intervals (CIs), were computed utilizing Stata 15.0. A forest plot was used for graphical representation, complemented by the construction of a summary receiver operating characteristic (sROC) curve to determine the area under the curve (AUC). Publication bias was evaluated using Deeks' funnel plot asymmetry test. All hypothesis tests were deemed statistically significant at a two-sided P-value threshold of less than 0.05. Additionally, to mitigate the risks of type I and II errors and to estimate the requisite sample size for robust meta-analysis, trial sequential analysis (TSA) was performed using TSA software (version 0.9.5.10 Beta). The TSA was configured with a power threshold of 80% and a bilateral significance level (α) of 0.05.

Results

Study selection

A comprehensive retrieval process yielded a total of 521 pertinent literature entries. Following a stringent screening based on predefined inclusion and exclusion criteria, 16 articles [12, 13, 15, 17–29] were selected for inclusion. These comprised six retrospective and ten prospective studies, together encompassing 3,409 patients. The literature screening process and outcomes are illustrated in Fig. 1.

Study characteristics

Of the included studies, twelve focused on the detection of pulmonary nodules measuring ≤ 3 cm, while four addressed nodules of mixed size. The distribution within the selected articles included seven studies on solitary pulmonary nodules (SPN), two on ground glass nodules (GGN), and seven on multiple-type nodules. Different methodologies for CTC capture were applied: nine studies employed negative enrichment (NE) combined with a folate receptor-targeting oligonucleotide (FRTO) probe, four utilized NE with a centromere of chromosome 8 (CEP8) probe, and three incorporated alternative techniques, including: (i) telomerase reverse transcriptase-based CTC detection (TBCD) combined with positive enrichment (PE) based on epithelial cell adhesion molecule (EpCAM), (ii) Ficoll-Hypaque density medium (FHDM) combined with DNA probe, and (iii) size enrichment (SE) combined with PE (EpCAM). Quantification methods for CTCs included fluorescence in situ hybridization (FISH) in five studies, ligand-targeted polymerase chain reaction (LT-PCR) in six, LT-PCR combined with Real-Time PCR (RT-PCR) in three, a combination of FISH and FlowSight in one study, and CELLTRACKSII in another. A detailed summary of the characteristics of these studies is presented in Table 1.

Quality assessment and risk of bias

The evaluation of the literature quality, conducted using the QUADAS-2 scale, is illustrated in Supplementary Fig. 1.

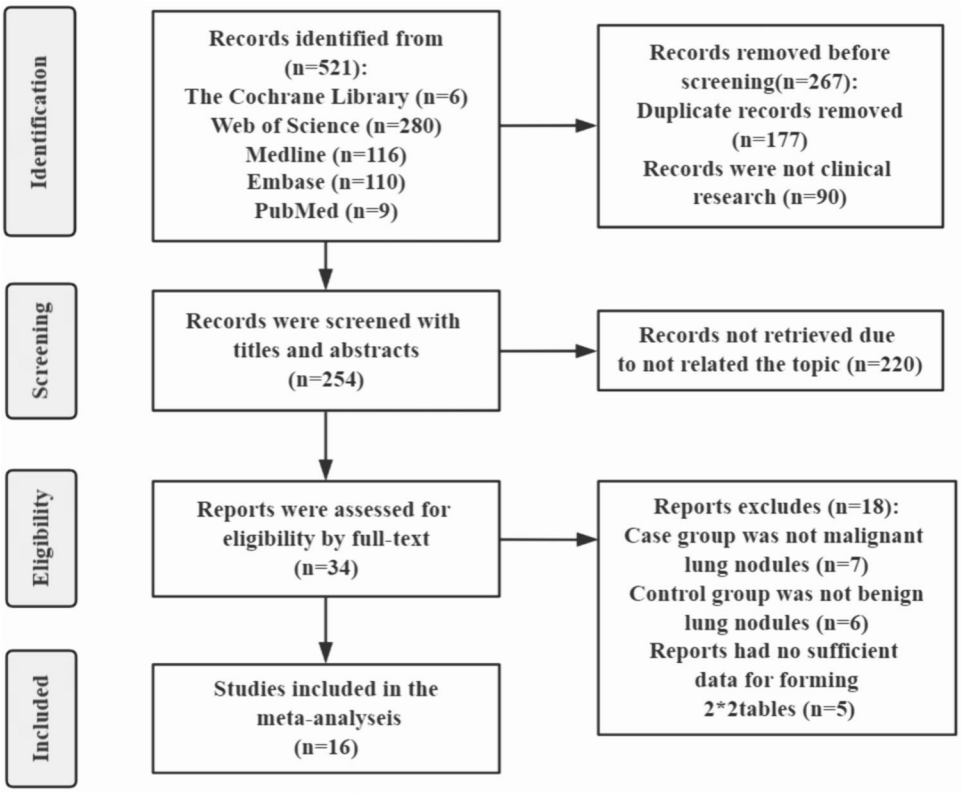


Fig. 1 Flow diagram of the study selection process and result

This assessment indicates that the overall quality of the included studies is high, lending credibility to the synthesized outcomes.

Accuracy of CTCs for pulmonary nodules

The threshold analysis yielded a Spearman correlation coefficient of 0.148, with an associated *P*-value of 0.403, exceeding the threshold of 0.05. This result suggests the absence of a threshold effect. The analysis of combined sensitivity and specificity produced I^2 values of 83.94% and 69.72%, respectively, indicating significant heterogeneity, thereby necessitating the adoption of subgroup analysis.

Pooled estimates of the diagnostic performance of CTCs include Sensitivity (0.84, 95% CI 0.80 to 0.87), Specificity (0.80, 95% CI 0.73 to 0.86) (Fig. 2), and AUC (0.89, 95%CI 0.86 to 0.92) (Fig. 3). Additionally, pooled diagnostic metrics were calculated as follows: PLR (4.23, 95% CI 3.12 to 5.72), NLR (0.20, 95% CI 0.16 to 0.25) (Supplementary Fig. 2), DOR (20.92, 95% CI 13.52 to 32.36), and diagnostic score (3.04, 95% CI 2.60 to 3.48) (Supplementary Fig. 3). The likelihood ratio scatter plot reveals that all pooled estimates and their corresponding 95% CIs are situated in the lower right quadrant, suggesting a need for caution when utilizing CTCs in the clinical diagnosis of benign and malignant pulmonary nodules (Supplementary Fig. 4).

Subgroup analysis

Substantial variations in diagnostic accuracy were identified across subgroups differentiated by lung nodule size, type, CTC capture technology, CTC quantification methodology, and application type. Notably, mixed-size type CTCs exhibited significantly enhanced diagnostic accuracy (DOR = 57.58, 95% CI 23.62 to 140.39) in comparison to pulmonary nodules ≤ 3 cm in size (DOR = 11.35, 95% CI 8.21 to 15.70). GGN CTCs showed higher diagnostic accuracy (DOR = 35.18, 95% CI 19.92 to 62.15) relative to SPN (DOR = 8.45, 95% CI 5.71 to 12.50) and subgroups with multiple types of confounding factors (DOR = 23.88, 95% CI 13.84 to 41.20).

In the specific domain of pulmonary nodule diagnosis, the most substantial diagnostic accuracy was achieved using FHDM + DNA probe capture technology (DOR = 787.03, 95% CI 99.14 to 6247.68). Among other CTC capture methods, significantly greater diagnostic accuracy was observed with SE + PE (EpCAM) (DOR = 38.71, 95% CI 7.31 to 204.90) and TBCD + PE (EpCAM) (DOR = 32.43, 95% CI 16.97 to 61.99), as opposed to NE + FRTO probe (DOR = 12.21, 95% CI 7.79 to 19.14) and NE + CEP8 probe (DOR = 11.36, 95% CI 6.76 to 19.08) techniques. Furthermore, diagnostic accuracy significantly increased when CTCs were combined with tumor–macrophage fusion cells (TMF) technology (DOR = 26.00, 95% CI 8.21

Table 1 Main features of the included studies

Study	Region	Simple size	Tp	Fp	Fn	Tn	Nodule type	Nodule size	CTC capture method	CTC quantitative method	Application type
Zhou [13]	China	271	150	36	41	44	SPN	≤3 cm	NE + FRTO probe	LT-PCR + RT-PCR	CTC
			136	15	55	65					Model
Yue [26]	China	65	38	4	5	18	GGN	Mixed	NE + FRTO probe	LT-PCR	CTC
			35	1	8	21					Model
Xu [25]	China	777	564	32	89	92	Multiple	Mixed	NE + FRTO probe	LT-PCR + RT-PCR	CTC
		264	189	10	16	49					CTC + TM
Wang [12]	China	299	190	7	85	17	SPN	≤3 cm	NE + FRTO probe	LT-PCR	CTC
			197	5	78	19					Model
Wan [24]	China	82	34	16	12	20	SPN	≤1 cm	NE + FRTO probe	LT-PCR + RT-PCR	CTC
			34	12	12	24					Model
Manjunath [23]	US	37	25	0	4	8	Multiple	Mixed	SE + PE(EpCAM)	CELLTRACKSII	CTC
			26	2	3	6					CTC + TMF
Ma [21]	China	224	108	2	63	51	Multiple	≤3 cm	NE + CEP8probe	FISH	CTC
			137	14	34	39					CTC + TM
Li [18]	China	190	92	10	41	47	Multiple	≤3 cm	NE + FRTO probe	LT-PCR	CTC
Zhong [29]	China	18	9	0	3	6	SPN	≤3 cm	NE + CEP8 probe	FISH	CTC
			10	0	2	6					CTC + TM
			9	0	3	6					CTC + TM
			10	1	2	5					CTC + TM
Zhang [27]	China	120	76	5	13	26	GGN	≤2 cm	TBCD + PE(EpCAM)	FISH + FlowSight	CTC
		131	94	2	28	7		≤1 cm			CTC
		89	59	3	8	19		1-2 cm			CTC
		120	80	5	9	26		≤2 cm			Model
Liu [20]	China	223	164	3	20	36	SPN	≤3 cm	NE + FRTO probe	LT-PCR	CTC
Liu [19]	China	53	38	6	3	6	Multiple	≤3 cm	NE + CEP8 probe	FISH	CTC
			39	6	2	6					Model
Zheng [28]	China	131	57	20	28	26	SPN	≤3 cm	NE + CEP8 probe	FISH	CTC
			49	6	36	40					CTC + TM
			79	23	6	23					CTC + TM
			66	13	19	33					Model
Katz [17]	US	89	37	0	9	43	Multiple	Mixed	FHDM + DNA probe	FISH	CTC
		118	58	0	3	57					
Zhou [15]	China	122	77	9	16	20	SPN	≤3 cm	NE + FRTO probe	LT-PCR	CTC
			52	5	41	24					CTC + TM
Ma [22]	China	224	137	14	34	39	Multiple		NE + FRTO probe	LT-PCR	CTC + TM

TP True-positive, *FP* False-positive, *FN* False-negative, *TN* True-negative, *SPN* Solitary pulmonary nodule, *GGN* Ground glass nodules, *Multiple* included multiple types of pulmonary nodules, *Mixed* included multiple sizes of pulmonary nodules, *NE* Negative enrichment, *FRTO* folate receptor-targeting oligonucleotide, *SE* Size enrichment, *PE(EpCAM)* positive enrichment based on epithelial cell adhesion molecule, *CEP8* centromere of chromosome 8, *TBCD* telomerase reverse transcriptase-based CTC detection, *FHDM* Ficoll-Hypaque density medium, *DNA* Included 3q29, 3p22.1, CEP10 and 10q22.3, *LT-PCR* The proprietary ligand-targeted polymerase chain reaction, *RT-PCR* Real-Time PCR, *FISH* fluorescence in situ hybridization, *TM* Tumor maker, *Model* model including CTC, clinical characteristics and biomarkers, *TMF* tumor-macrophage fusion cells

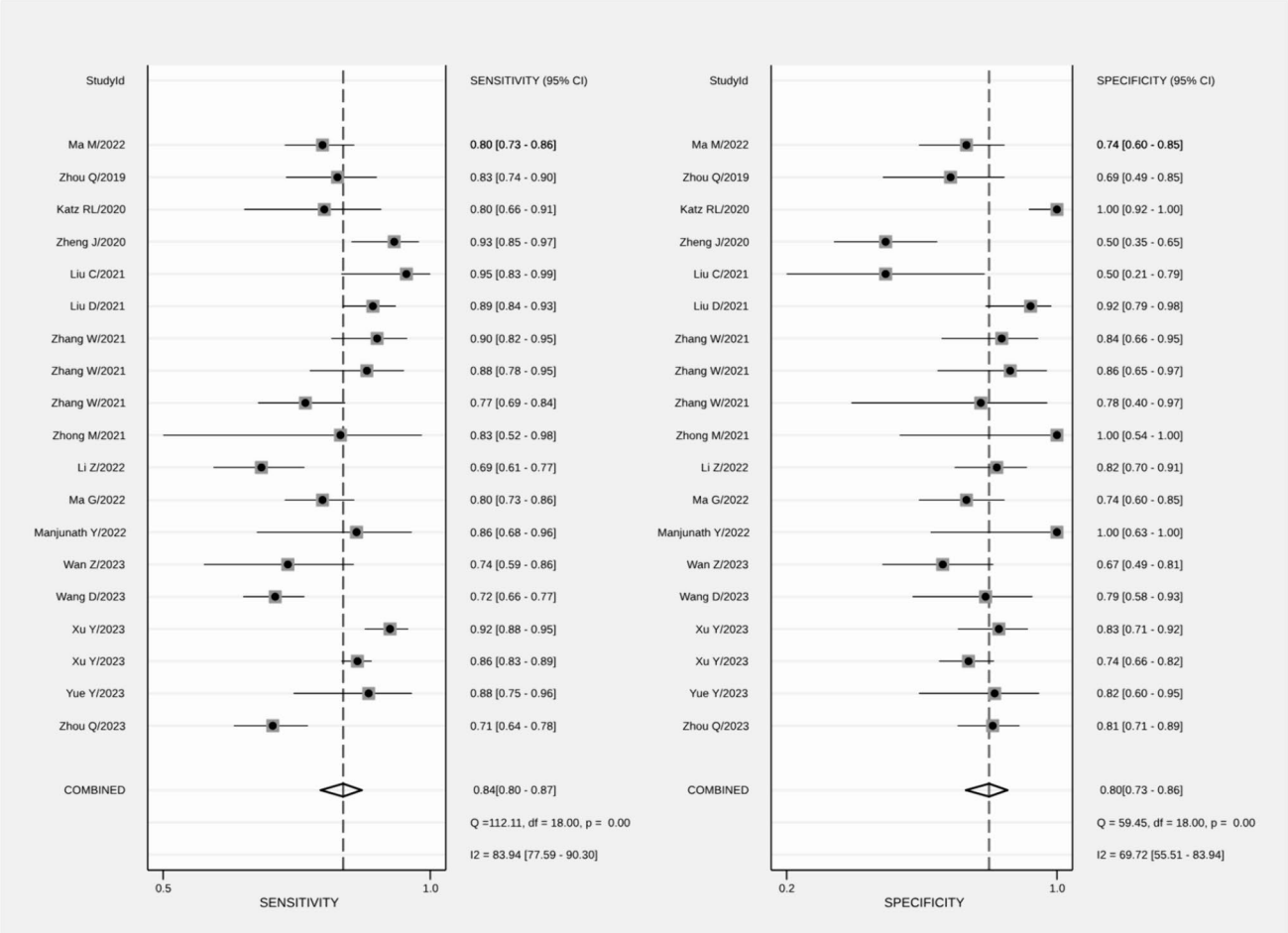


Fig. 2 Forest plots of pooled sensitivity and specificity for circulating tumor cells (CTCs) in the diagnosis of benign and malignant pulmonary nodules

to 82.37) compared to methodologies utilizing only CTCs (DOR = 17.55, 95% CI 9.86 to 31.21), CTCs combined with tumor markers (TM) (DOR = 14.66, 95% CI 8.59 to 25.01), and models (DOR = 13.14, 95% CI 7.42 to 23.24) (Table 2).

Sensitivity analysis and publication bias

Deek's funnel plot analysis, as presented in Supplementary Fig. 5, indicated a slope coefficient exceeding 0.05, suggesting the absence of publication bias among the studies included. This finding supports the robustness of the compiled data, reinforcing the credibility of the overall analysis outcomes.

Clinical application value

To ascertain the clinical relevance, a simulated clinical setting was employed, assigning a predicted probability of 50% for the PLR and 20% for the NLR. Furthermore, the simulation allocated a posterior probability of 81% for a positive test outcome and 17% for a negative result, as depicted in Fig. 4. This approach enhances the interpretability of diagnostic efficiency within a clinical framework.

Sequential analysis of trials

The results from the TSA are detailed in Fig. 5. This meta-analysis incorporated sixteen studies, establishing a required sample size (RIS) of 1483. The analysis revealed that the cumulative Z value exceeded both the conventional significance threshold of $Z = 1.96$ and the TSA-specific threshold. This indicates that a definitive conclusion about the efficacy of CTCs as a biomarker was reached before attaining the anticipated sample size. The TSA substantiates the utility of low levels of CTCs as a significant marker for the exclusion of benign pulmonary nodules, thereby affirming their diagnostic value in clinical settings.

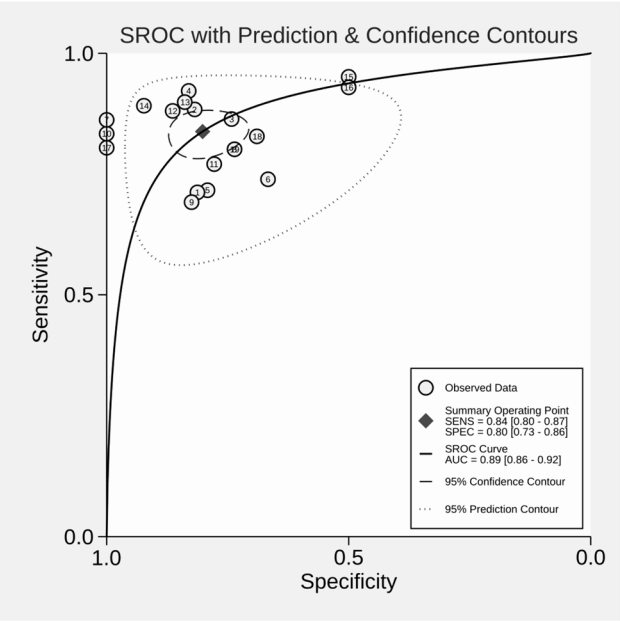


Fig. 3 SROC curve plot of CTCs in the diagnosis of benign and malignant pulmonary nodules

Discussion

This systematic review and meta-analysis represents an inaugural effort to appraise the diagnostic efficacy of CTCs in differentiating between benign and malignant pulmonary nodules. Prior investigations into this subject have predominantly suffered from inadequate statistical power and have been restricted to analyses involving diminutive cohorts [23, 29]. In an endeavor to facilitate a more comprehensive evaluation, data from extant publications were aggregated, incorporating 35 datasets from 16 distinct studies, encompassing a total of 2717 cases and 810 control subjects. The findings decisively demonstrate that CTCs exhibit heightened sensitivity and specificity in the differential diagnosis of pulmonary nodules, thereby providing substantial reinforcement for the clinical adjudication of such conditions. Additionally, the results from the TSA affirm that CTCs furnish compelling evidence for the exclusion of benign pulmonary nodules. This mitigates the prevalent limitations associated with the high rate of false positives encountered in LDCT screenings, thus aiding in the more precise diagnosis of pulmonary nodules.

Table 2 Results of subgroup analysis

Classification	N	Sensitivity			Specificity			DOR		
		Value(95%CI)	I^2(%)	P	Value (95%CI)	I^2(%)	P	Value(95%CI)	I^2(%)	P
Nodule size										
≤3cm	27	0.76(0.74,0.77)	82.8	0.01	0.74(0.71,0.77)	74.2	0.01	11.35 (8.21,15.71)	59.7	0.01
Mixed	8	0.88(0.86,0.90)	45.3	0.08	0.86(0.82,0.89)	85.4	0.01	57.58(23.62,140.39)	65.0	0.01
Nodule type										
SPN	17	0.74(0.72,0.76)	81.4	0.01	0.71(0.67,0.74)	75.2	0.01	8.45(5.71,12.50)	57.9	0.01
GGN	6	0.84(0.81,0.88)	40.5	0.14	0.85(0.78,0.91)	0.0	0.69	35.18(19.92,62.15)	0.0	0.71
Multiple	12	0.83(0.81,0.85)	88.4	0.01	0.82(0.79,0.85)	85.4	0.01	23.88(13.84,41.20)	63.2	0.01
CTC capture method										
NE + FRTO probe	15	0.79(0.77,0.81)	89.2	0.01	0.75(0.72,0.78)	69.4	0.01	12.21(7.79,19.14)	75.9	0.01
NE + CEP8 probe	12	0,75(0.72,0.76)	84.2	0.01	0.73(0.68,0.76)	81.6	0.01	11.36(6.76,19.08)	46.2	0.04
TBCD + PE(EpCAM)	4	0.84(0.80,0.88)	60.3	0.06	0.84(0.75,0.91)	0.0	0.10	32.43(16.97,61.99)	0.0	0.55
SE + PE(EpCAM)	2	0.88(0.77,0.95)	0.0	0.69	0.88(0.62,0.98)	67.3	0.08	38.71(7.31,204.90)	0.0	0.47
FHDM + DNA probe	2	0.89(0.81,0.94)	82.5	0.02	1.00(0.96,1.00)	0.0	1.00	787.03(99.14,6247.68)	0.0	0.41
CTC quantitative method										
FISH	14	0.77(0.74,0.80)	85.0	0.01	0.79(0.75,0.83)	88.6	0.01	17.78(8.97,35.24)	68.2	0.01
LT-PCR	9	0.75(0.73,0.77)	86.8	0.01	0.81(0.76,0.85)	41.9	0.09	13.95(7.94,24.51)	62.1	0.01
LT-PCR + RT-PCR	6	0.83(0.81,0.85)	88.6	0.01	0.71(0.66,0.75)	78.3	0.01	10.13(4.74,21.67)	86.4	0.01
FISH + FlowSight	4	0.84(0.80,0.88)	60.3	0.06	0.84(0.75,0.91)	0.0	0.95	32.43(16.97,61.99)	0.0	0.55
CELLTRACKSII	2	0.88(0.77,0.95)	0.0	0.69	0.88(0.62,0.96)	67.3	0.08	38.71(7.31,204.90)	0.0	0.47
Application type										
CTC	18	0.80(0.78,0.81)	85.5	0.01	0.78(0.75,0.81)	86.6	0.01	17.55 (9.86, 31.21)	79.4	0.01
Model	7	0.76(0.73,0.79)	78.9	0.01	0.77(0.72,0.82)	58.3	0.03	13.14 (7.42,23.24)	49.8	0.06
CTC + TM	9	0.79(0.77,0.82)	90.5	0.01	0.76(0.71,0.81)	71.0	0.01	14.66 (8.59,25.01)	49.7	0.04
CTC + TMF	1	0.90(0.81,0.95)	0.0	1.00	0,75(0,53,0.90)	0.0	1.00	26.00 (8.21,82.37)	0.0	1.00

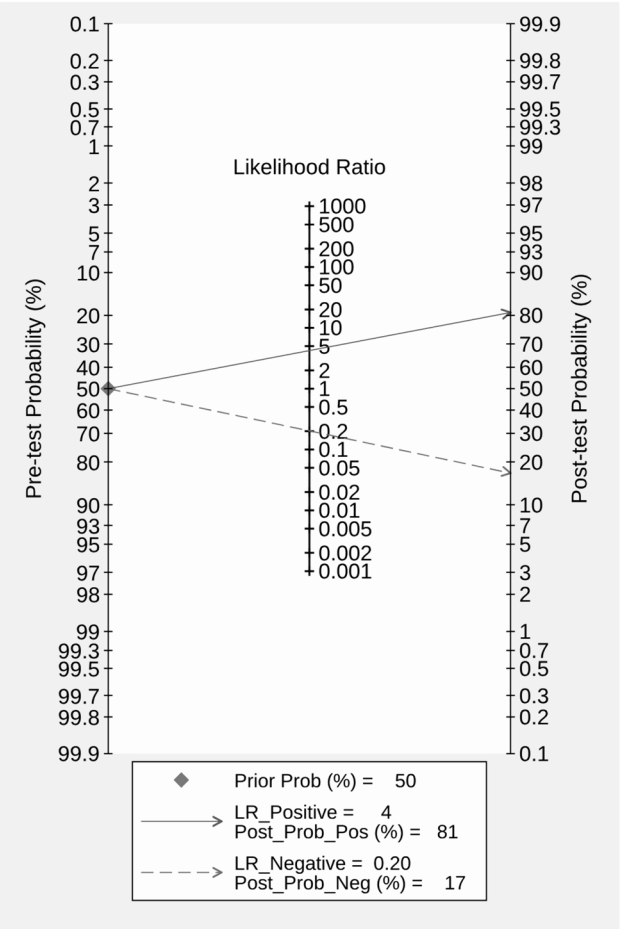


Fig. 4 Fagan nomogram of accuracy for CTCs diagnosis of benign and malignant pulmonary nodules

During the initial phases of primary tumor development in malignancies, such as lung, breast, and colon cancer, cells are shed into the bloodstream. These cells, termed CTCs, can be isolated and identified using various techniques that exploit their unique physical and biological properties. The analysis of CTCs is regarded as a real-time "liquid biopsy" that allows for the ongoing monitoring of cancer patients [30]. Liquid biopsy offers several advantages, including its minimally invasive nature, cost-effectiveness, and high efficiency. Research has demonstrated that patients with Chronic Obstructive Pulmonary Disease (COPD), who are monitored annually for CTC positivity, often undergo surgical excision of lung nodules detected in CT scans between one and four years following CTC detection. These procedures frequently result in a histopathological confirmation of early-stage lung cancer [31]. These findings highlight the potential utility of CTCs in the screening and early detection of lung cancer. Furthermore, researchers have employed high-throughput targeted DNA methylation sequencing of circulating tumor DNA (ctDNA) to differentiate between benign and malignant lung nodules smaller than 3 cm, achieving a sensitivity and specificity of 79.5% (95% CI 63.5 to 90.7) and 85.2% (95% CI 66.3 to 95.8), respectively [32]. The markedly low abundance of ctDNA in early-stage lung cancer, coupled with the required technical sophistication, significantly restricts the practicality of this diagnostic modality for routine clinical application [33].

Subgroup analysis by nodule size revealed superior diagnostic performance of CTCs for mixed-size groups compared to nodules ≤ 3 cm. Studies have shown a correlation between the size of pulmonary nodules and their benign or malignant nature, and a larger nodule volume is generally associated with an increased probability of malignancy [34]. Nodule volume is an important parameter to evaluate pulmonary nodule properties. Pulmonary nodules with a volume ≥ 300 mm^3 or a diameter ≥ 1 cm are classified into a subgroup with uncertain benign or malignant nature, necessitating further diagnostic evaluation [35]. Additionally, one study reported that the average diameter and volume of lung nodules possess comparable diagnostic utility in the identification of malignant tumors [36]. This study demonstrated that CTCs had a sensitivity of 0.76 (95% CI 0.74 to 0.77) and a specificity of 0.74 (95% CI 0.71 to 0.77) in diagnosing pulmonary nodules ≤ 3 cm, and a sensitivity of 0.76 (95% CI 0.69 to 0.81) in diagnosing nodules ≤ 1 cm. This was significantly superior to LDCT screening alone [37].

Subgroup analysis delineating nodule types revealed that CTCs exhibited the highest diagnostic accuracy in the assessment of GGNs, achieving a sensitivity of 0.84 (95% CI 0.81 to 0.88) and a specificity of 0.85 (95% CI 0.78 to 0.91). These outcomes were markedly superior to those observed for SPNs and multiple nodule types, suggesting that CTCs may offer enhanced diagnostic efficacy, particularly for GGNs, potentially elucidating underlying confounding factors. First, the superior diagnostic outcomes for GGNs could be attributed to variations in nodule size; the GGN group encompasses nodules of various dimensions, whereas SPNs are characterized as nodules ≤ 3 cm [38]. Second, the association might be linked to the technological modalities employed for CTC detection. In the studies reviewed, TBCD combined with PE (EpCAM) technology was utilized for CTC capture in four datasets specifically for GGN diagnosis, with subsequent CTC quantification conducted using FISH and FlowSight technologies. These methodologies have demonstrated robust diagnostic performance, particularly in the identification of GGNs.

The subgroup analysis focusing on the methods of CTC capture indicated varied levels of diagnostic efficacy across different technologies. Specifically, the FHDM + DNA probe technology displayed superior diagnostic performance compared to other methods; however, the analysis was limited by data instability due to a small dataset. This

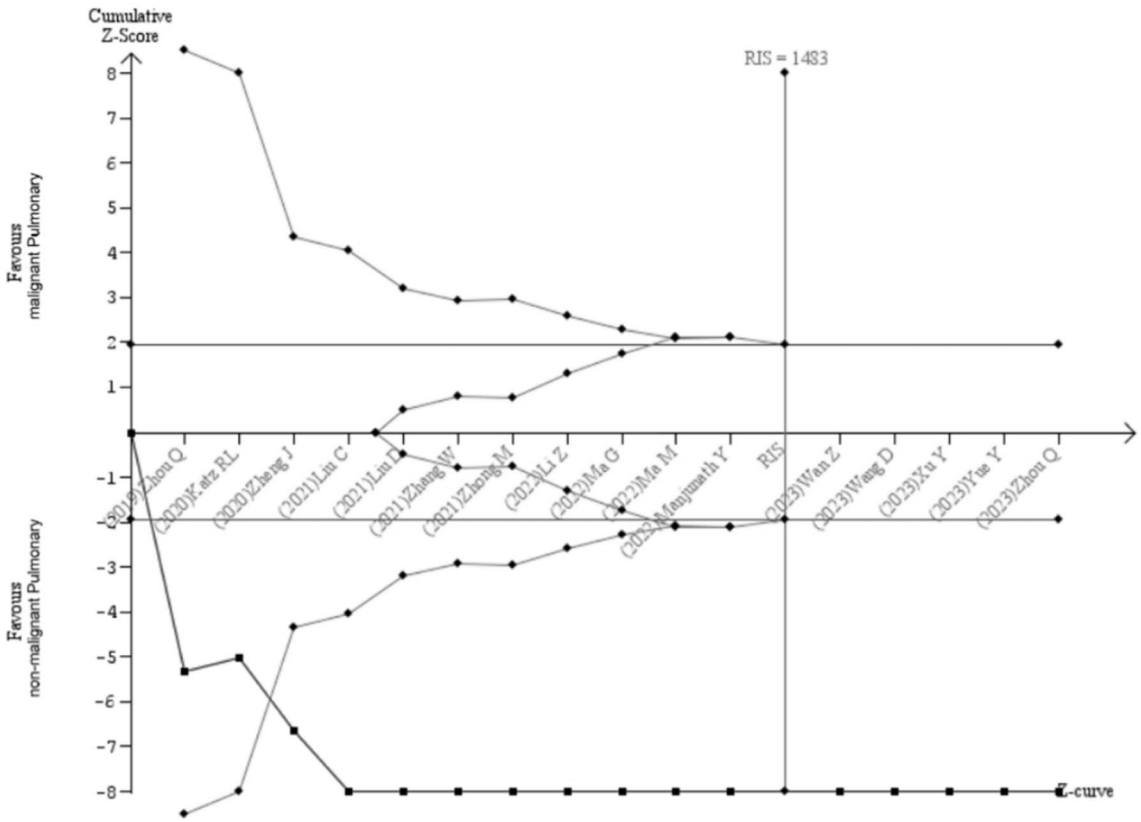


Fig. 5 The *Z*-curve plot of the efficacy of CTCs in the diagnosis of benign and malignant pulmonary nodules

highlights a need for caution in interpreting these results as definitive. The SE + PE (EpCAM) technique also showed impressive diagnostic capabilities, with both sensitivity and specificity at 0.88 (95% CI 0.77 to 0.95 and 0.62 to 0.98, respectively). However, this method was used exclusively with the CELLTRACKSII quantification technique, which limits the generalizability of the findings due to the narrow data scope. Moreover, the TBCD + PE (EpCAM) technique demonstrated good diagnostic outcomes, with a sensitivity of 0.84 (95% CI 0.80 to 0.88) and a specificity of 0.84 (95% CI 0.75 to 0.91). In all studies that utilized this method, CTC quantification was performed using a combination of FISH and FlowSight technologies, suggesting that this combination might be particularly effective. The comparative accuracy of PE capture techniques (AUC = 0.9198) was notably higher than that of NE (AUC = 0.8478), as shown in the subgroup analysis. This underscores the effectiveness of PE-based capture techniques in isolating CTCs with high precision [39] (Fig. 6).

In the analysis of quantitative methods for CTCs, the combined use of LT-PCR and RT-PCR showed higher sensitivity (0.83, 95% CI 0.81 to 0.85) than LT-PCR alone, suggesting that a combination of quantitative techniques can enhance the sensitivity of CTC detection. In clinical settings, the choice between these quantitative techniques should be guided by the specific diagnostic outcomes desired. Using both techniques concurrently might improve the accuracy of results, particularly when employing the NE + FRTO probe for CTC capture. This comprehensive approach could provide a more reliable framework for CTCs-based diagnostics, improving the precision and utility of these methods in clinical practice.

In a subgroup analysis stratified by application type, a slight enhancement in diagnostic accuracy was observed with CTCs (AUC = 0.8771) for the detection of pulmonary nodules when compared to a standard model (AUC = 0.8569) and a CTC + TM (AUC = 0.8601) as depicted in Supplementary Fig. 6. This outcome notably challenges the prevalent assumption that integrating biomarkers with other clinical parameters enhances diagnostic efficacy more than their solitary use [13]. While this finding introduces some uncertainty, it notably suggests that CTCs alone may offer superior diagnostic potential for identifying pulmonary nodules. The deployment of advanced CTC detection technologies, which facilitate the precise capture and quantification of CTCs, could potentially obviate the need to combine these cells with additional clinical markers, thus simplifying and accelerating their clinical application. Furthermore, while

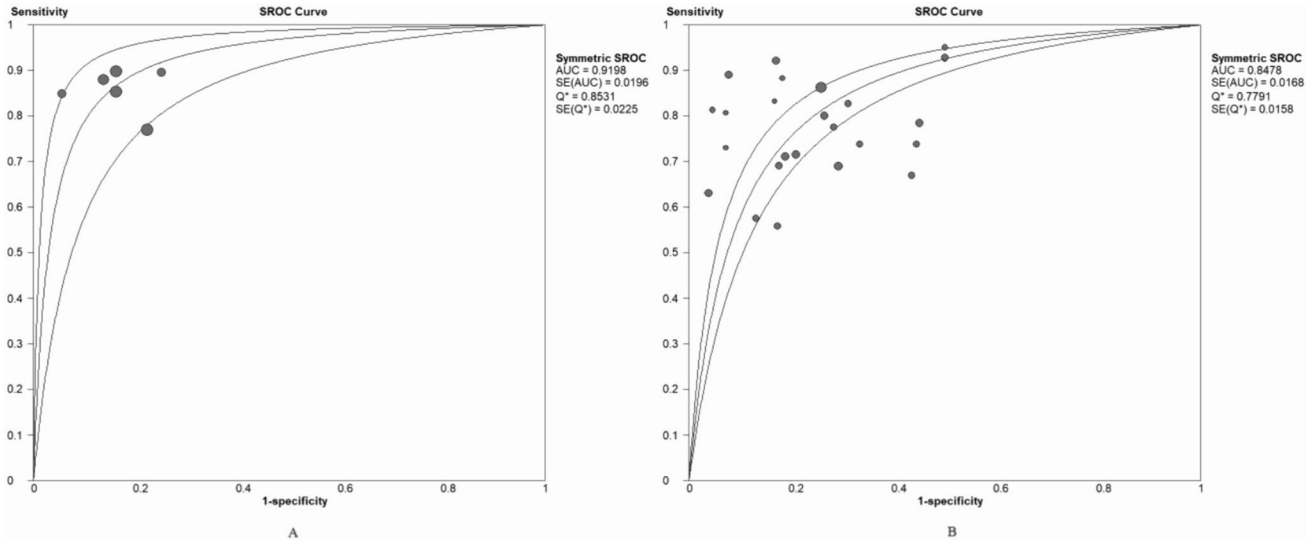


Fig. 6 SROC curve of different CTCs capture techniques in the diagnosis of benign and malignant pulmonary nodules. **A** SROC curve of PE capture techniques. **B** SROC curve of NE capture techniques

LDCT is recommended for early lung cancer detection in high-risk populations, it is plagued by a high false-positive rate [40]. In contrast, serum CTC detection, as a minimally invasive and convenient diagnostic approach, is likely to gain greater acceptance compared to other radiation-based imaging techniques. However, significant regional and institutional disparities exist in the implementation of CTC testing. In settings endowed with abundant resources, the deployment of advanced CTC detection technologies is often feasible. Conversely, in environments constrained by limited resources, the high costs associated with this equipment may impede its widespread utilization. Currently, fluorodeoxyglucose (18F-FDG) positron emission tomography–computed tomography (PET-CT) is endorsed for the diagnosis and staging of non-small cell lung cancer (NSCLC), although the accuracy of this diagnostic tool has been subject to scrutiny [41]. In regions with restricted resources, PET-CT remains a prevalent initial diagnostic approach for evaluating larger pulmonary nodules. Despite this, CTC testing, which offers a non-invasive and potentially more precise alternative for early cancer detection, has yet to be widely integrated into clinical practice.

This study possesses several limitations that warrant consideration. First, the analysis did not differentiate among types of malignant pulmonary nodules, such as small-cell lung cancer, lung adenocarcinoma, and lung squamous cell carcinoma. This limitation stems from the categorization used within the datasets analyzed. Second, there was no accommodation for classifying and analyzing pulmonary nodules based on their sizes despite the well-documented correlation between nodule size and the likelihood of malignancy. Lastly, the absence of a standardized cut-off value, attributable to the novel nature of the detection techniques utilized, may have impacted the precision of our findings. This lack of standardization highlights the need for further validation and refinement of these diagnostic methods to enhance their accuracy and clinical relevance.

Conclusion

In conclusion, CTCs exhibit promising diagnostic accuracy in distinguishing between benign and malignant pulmonary nodules, thereby significantly enhancing the diagnostic process of nodules detected through LDCT screening. Nevertheless, the current landscape of CTC detection techniques is characterized by variability in technology platforms and analytical methodologies, resulting in divergent levels of sensitivity and specificity. Future investigations must prioritize the standardization of these technologies to ensure consistency and reproducibility across diverse laboratory and clinical settings. Furthermore, there is a pressing need to advance novel methodologies, such as integrating biomarker screening with machine learning algorithms, to augment the precision and efficiency of CTC detection. Advancements in these realms will facilitate more precise identification and quantification of CTCs, thereby enabling the execution of prospective studies to validate their performance across varied sizes, volumes, and subtypes of lung nodules. Such strides are essential for determining the most efficacious approaches for clinical implementation.

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Ethical approval Institutional Review Board approval was not required because this article is a review.

Informed consent Written informed consent was not required for this study because this article is a review.

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